Benjamin D. Killeen

Contact

Computer Aided Medical Procedures
Technical University of Munich
Boltzmannstr. 3
85748 Garching, Germany

Voice: +49 152 92608722
Office: CIT 03.13.044
Mail: bd.killeen@tum.de
Web: benjamindkilleen.com

Summary

As a postdoctoral researcher at the Technical University of Munich, I research the future of AI- and robot-assisted healthcare, leveraging enormous data generated by health systems together with sophisticated simulation environments to improve patient outcomes. Every day, I aim to build community in the classroom, in my network, and through professional societies to foster an inclusive environment.

Education

Ph.D. in Computer Science 29.08.2019 – 24.03.2025
Johns Hopkins University
Primary advisor: Mathias Unberath
Secondary advisor: Gregory D. Hager
Thesis: *Intelligent Assistance Systems for X-ray Image-guided Interventions via Surgical Simulation Environments*

M.S.E. in Computer Science 29.08.2019 – 01.08.2022
Johns Hopkins University

B.A. in Computer Science with Honors 21.09.2015 – 15.06.2019
Minor in Physics
University of Chicago
Honors thesis advisor: Gordon Kindlmann

Professional Employment

Postdoc 15.09.2025 – now
Technical University of Munich, Chair for Computer Aided Medical Procedures
With Nassir Navab.

Founding Engineer / AI Consultant 05.01.2026 – now
Semaphor Surgical – Code that operates.

Computer Vision / AI Intern 15.06.2020 – 31.07.2020
Intuitive Surgical Inc.
Applied Research
With Omid Mohareri.

Software Development Intern 11.06.2018 – 24.08.2018
Epic Systems, Center for Cognitive Computing

Research Intern, 26.06.2017 – 31.08.2017
IBM Research - Almaden, Neuromorphic Computing
With Geoffrey Burr.

Publication Summary

Out of 38 publications, I have (first/co)-authored 7/10 journal articles, 6/6 conference papers, and 3/6 preprints, and I am an inventor on 3 patents or patent applications in process. A detailed list is available below and on Google Scholar.

Selected Awards

Personal Awards

5. **Siebel Scholar** 2024
   Recognizes top students at the world's leading graduate schools of business, computer science, bioengineering, and energy science.
4. Finalist, **WSE Excellence in Teaching, Advising, and Mentoring Award** 2024
   Honors JHU faculty and graduate students who excel in the arts of teaching, advising, and mentoring.
3. DAAD AI-Net Fellow, **Postdoc-NeT-AI Networking Week on Human-centered AI** 2023
   Postdoctoral networking tour in Germany supported by the German Academic Exchange Service (Deutscher Akademischer Austauschdienst).
2. Recipient, **Link Foundation Fellowship in Modeling, Simulation, and Training** 2023
   Two-year fellowship for Ph.D. students to fund their research.
1. **LCSR Fellowship for Outstanding Incoming Ph.D. Students** 2019

Publication Awards

7. **Best Paper Award** 2025
   For paper [J-14] at IPCAI 2025.
6. **Bench-to-Bedside Award** 2024
   For paper [J-7] at IPCAI 2024.
5. Finalist, **Best Paper Award** 2024
   For paper [J-9] at IPCAI 2024.
4. Honorable Mention, **Bench-to-Bedside Award** 2023
   For paper [J-5] at IPCAI 2023.
3. Runner Up, **Best Paper Award, Physics of Medical Imaging** 2022
   For paper [C-5] at SPIE Medical Imaging 2022.
2. **Best Paper Award in Bioengineering** 2021
   For paper [C-4] at IEEE BIBE 2021.
1. **Kaggle COVID-19 Dataset Award** 2020
   For our US county-level dataset described in [M-1].

Reviewer Awards

3. **IEEE TMI Distinguished Reviewer**, Bronze Level 2024
2. **IPCAI Outstanding Reviewer Award** 2024
1. Honorable Mention, **MICCAI Outstanding Reviewer Award** 2023

Funding

Fellowships and Small Awards

3. **MICCAI NIH Registration Grant**: USD 780 — 2025
2. **DAAD Postdoc-Net-AI Travel Grant**: USD 2,000 — 2023
1. **Link Foundation Fellowship in MS&T**: USD 70,000 — 2023
   "Interactive Digital Twins for Simulating the Future of Work in AI- and Robot-assisted Operating Rooms"

Service and Leadership

Societies

TRU-UE 197 Steward, Dept. of Computer Science — 2024

Social Events Officer, MICCAI Student Board — 10.2023 – 10.2024

President, LCSR Graduate Student Association — 08.2022 – 12.2023
Established an executive board managing $8,000/yr in student resources.

Sports Officer, MICCAI Student Board — 12.2021 – 09.2023
Organizer for athletic events at the MICCAI conference.
On-site representative and MICCAI Educational Challenge reviewer.

Head of Student Resources — 09.2020 – 08.2022
LCSR Graduate Student Committee

Academic Services

Seminar Course Assistant — 2023
Future Faculty: Preparing a New Generation of PIs for the Academic Job Search
Department of Computer Science, Johns Hopkins University

Organizer — 2023
Focus Group on Graduate Student Space
Laboratory for Computational Sensing and Robotics, Johns Hopkins Univ.

Brainlab Loop-X Trainer and Coordinator — 2022 – now
Laboratory for Computational Sensing and Robotics, Johns Hopkins Univ.

Robotorium and Mock OR Tours — 2022, 2023
Laboratory for Computational Sensing and Robotics, Johns Hopkins Univ.

Organizer

Co-organizer — 2026
Medical Foundation Models and Benchmarks
ECCV Workshop, Malmö, Sweden

Co-organizer — 2026
Summer School on AI, Robotics and XR for Clinical Applications
Shenzhen, China

Co-organizer — 2025
Medical Augmented Reality Summer School (MARSS)
Zurich, Switzerland

Co-organizer 2024
PENGWIN: Pelvic Bone Fragments with Injuries
MICCAI Challenge, Marrakesh, Morocco

Journal Reviewer

Nature Communications
IEEE Transactions on Medical Imaging (TMI)
Journal of Machine Learning Research (JMLR)
Quantitative Imaging in Medicine and Surgery (QIMS)
Journal of Visualized Surgery (JOVS)
IEEE Robotics and Automation Letters (RA-L)
Computer Assisted Surgery (CAI)
Nature Scientific Data
Medical Image Analysis (MedIA)

Conference Reviewer

Medical Image Computing and Computer Assisted Interventions (MICCAI)
International Conference on Information Processing in Computer-Assisted Interventions (IPCAI)
International Symposium on Medical Robotics (ISMR)
IEEE International Conference on Computer Vision (ICCV)
IEEE European Conference on Computer Vision (ECCV)
IEEE Asian Conference on Computer Vision (ACCV)
IEEE/CVF Computer Vision and Pattern Recognition (CVPR)

Teaching

Computer Integrated Surgery II (EN.601.456/656)
Department of Computer Science, Johns Hopkins University

Project mentor: *Diffusion-based Vertebra Reconstruction* 2025
Project mentor: *Measuring Variability of Pelvic Standard Views in Virtual Reality* 2024
Voted runner-up, **Best Project Award**.
Project mentor: *A Cannula Marker Body for Tracker-free Surgical Navigation during Kirschner Wire Placement* 2024
Project mentor: *Bringing View-invariant X-ray Image Analysis into the Operating Room* 2024
Project mentor: *Recreating Pelvic Trauma Surgery in Virtual Reality for the Development of Novel C-arm Interfaces* Spring 2023
Voted **Best Project Award**.
Project mentor: *Making 2D/3D Registration Accessible* 2023
Project mentor: *3D Segmentation of Hard and Soft Tissue for Simulating X-ray Image Formation with Deep Learning* 2022

Computer Integrated Surgery I (EN.601.455/655)
Department of Computer Science, Johns Hopkins University
Teaching assistant. Quality: 4.32/5.00 (sample size: 86) Fall 2022
Teaching assistant. Quality: 4.13/5.00 (sample size: 63) Fall 2021

Introduction to Computer Science (CMSC 14100/14200)
Department of Computer Science, University of Chicago
Course assistant Summer 2019

Machine Learning and Large Scale Data Analysis (CMSC 25025)
Department of Computer Science, University of Chicago
Teaching assistant Spring 2019

Scientific Visualization (CMSC 23710)
Department of Computer Science, University of Chicago
Course assistant Winter 2019

Supervision

As a postdoc at the Technical University of Munich, I directly supervise students' PhD projects, master's theses, and contributions to research.

5. **Tofig Rahimov**, Technical Univ. Munich 03.2026 – now
   Bacheslor's thesis project.
4. **Lucas Ronchetti**, Technical Univ. Munich 11.2025 – now
   Interdisciplinary research project.
3. **Ameer Hamza**, Technical Univ. Munich 10.2025 – now
   Visiting Ph.D student from **Kyung Hee University**.
2. **Miriam Senne**, Technical Univ. Munich 09.2025 – now
1. **Maximilian Fehrentz**, Technical Univ. Munich 09.2025 – now

As a PhD student at Johns Hopkins, I supervised students' contributions to research. Where known, career steps after completing their research effort are provided.

Select Graduate

16. **Florence Klitzner**, Johns Hopkins Univ. 03.2025 – 08.2025
    Visiting from the Technical University of Munich.
15. **Nidhi Balodi**, Johns Hopkins Univ. 09.2024 – 08.2025
    Visiting from the Technical University of Munich.
14. **Eric Song**, Johns Hopkins Univ. 01.2025 – 06.2025
13. **Iou-Sheng "Danny" Chang**, Johns Hopkins Univ. 02.2023 – 05.2023
12. **Ching-Yang "Austin" Huang**, Johns Hopkins Univ. 02.2023 – 05.2023
    Austin joined **XtalPi** as an **Automation Engineer**.
11. **Yuxuan Zhao**, Johns Hopkins Univ. 02.2023 – 05.2023
10. **Xu "Lance" Lian**, Johns Hopkins Univ. 09.2023 – 12.2023
9. **Bohua Wan**, Johns Hopkins Univ. 06.2023 – 01.2025
   Bohua completed [M-4] while a junior PhD student at JHU.
8. **Hengyu Cao**, Johns Hopkins Univ. 08.2023 – 12.2023
7. **Shreayan Chaudhary**, Johns Hopkins Univ. 05.2023 – 05.2024
   Joined **Seagate Technology as a Machine Learning Engineer**.

6. **Han Zhang**, Johns Hopkins Univ. 01.2023 – 12.2023
Joined **Johns Hopkins University as a Ph.D. Student**.
5. **Zixuan Liu**, Johns Hopkins Univ. 01.2023 – 09.2023
Joined **Vanderbilt University as a Ph.D. Student**.
4. **Aditya Kulkarni**, Johns Hopkins Univ. 09.2022 – 01.2025
Joined **St. Jude Children's Hospital** as a **CV/ML Researcher**.
3. **Qiyuan Wu**, Johns Hopkins Univ. 08.2022 – 06.2023
Joined **Cornell University as a Ph.D. Student**.
2. **Zidi Tao**, Johns Hopkins Univ. 10.2021 – 06.2022
Joined **Rensselaer Polytechnic Institute as a Ph.D. Student**.
1. **Shreaya Chakraborty**, Johns Hopkins Univ. 08.2020 – 09.2021
Joined **PathAI as a Machine Learning Engineer**.

Select Undergraduates

11. **Sampath Rapuri**, Johns Hopkins Univ. 11.2024 – now
10. **Jeremy Ko**, Johns Hopkins Univ. 08.2024 – 08.2025
9. **Janya Budaraju**, Johns Hopkins Univ. 02.2024 – 01.2025
Recipient of the Pistritto Fellowship based on her undergraduate research.
Will join **Intuitive Surgical**.
8. **Samhith Bhrugubanda**, Johns Hopkins Univ., 02.2024 – 11.2024
7. **Asmitha Sathya**, Johns Hopkins Univ., 09.2023 – 12.2023
6. **Darren Shih**, Johns Hopkins Univ. 09.2023 – 12.2023
5. **William "Liam" Wang**, Johns Hopkins Univ. 01.2023 – 12.2023
Joined the **University of Michigan** as a **Ph.D. Student.**
Fellow in the **NSF Graduate Research Fellowship Program**.
4. **Sambhav Chordia**, Johns Hopkins Univ. 06.2022 – 12.2022
3. **Sean Sebastian Darcy**, Johns Hopkins Univ 10.2021 – 10.2022
Joined the **California Institute of Technology as a Ph.D. Student**.
2. **Nethra Venkatayogi**, Johns Hopkins Univ. 05.2021 – 10.2021
Visiting from the University of Texas at Austin.
Joined **Johns Hopkins University as a Ph.D. Student**.
1. **Max Judish**, Johns Hopkins Univ. 01.2021 – 08.2021
Visiting from Brown University.

TALKS AND PRESS

Invited Talks and Demos

15. Special Colloquium Lecture 12.01.2026
STIMULATE Research Center, Magdeburg, Germany
"In Silico Models for AI-Driven Surgical Teaming"

14. Master's Seminar Invited Lecture 11.12.2025
Technical University of Munich, Munich, Germany
"Foundation Models for Minimally Invasive Image-guided Surgery"

13. Invited Talk 28.10.2024
Intuitive Surigcal, inc.
"Building Surgical AI One Simulation at a Time"

12. End of Semester Social **Selected Posters and Demos** 14.05.2024
Data Science and AI Runner up, **Audience Choice Award**.

11. Invited Demonstration 09.2024
Nemertes [Next] Conference
"Simulation-driven AI in the Operating Room"

10. Engineering Innovation Special Lecture 07.2024
Whiting School of Engineering, Johns Hopkins University, Baltimore, USA
"Building Surgical AI One Simulation at a Time"

9. AIMS Lecture Series 06.2024
Institute for Artificial Intelligence in Medicine (IKIM), Essen, Germany
"Advancing Interventional Healthcare One Simulation at a Time"

8. End of Semester Social, **Selected Posters and Demos** 05.2024 Johns Hopkins Data Science and AI Institute, Baltimore, USA
"Neural Digital Twins"

7. Malone Center Research Lunch 05.2024
Malone Center for Engineering in Healthcare, Baltimore, USA
"Advancing Interventional Healthcare One Simulation at a Time"

6. Malone Center Trainee Mix and Mingle 04.2024
Malone Center for Engineering in Healthcare, Baltimore, USA
"The Future of Simulation-Driven Interventional Healthcare"
Runner up, **Audience Choice for Best Presentation**.

5. IHU Seminar Series 03.2024
Institute of Image-guided Surgery (IHU), Strasbourg, France
"Advancing Interventional Healthcare One Simulation at a Time"

4. CAMP Seminar Series 03.2024
In the Postdoc-NeT-AI Networking Week on Human-centered AI
Technical University of Munich, Munich, Germany
"Advancing Interventional Healthcare One Simulation at a Time"

3. medPhoton Invited Talk Series — 06.2023
   medPhoton, Salzburg, Austria
   "Robotic X-ray Imaging Interfaces"

2. FDA DIDSR Seminar Series — 05.2023
   Food and Drug Administration (FDA), Silver Spring, MD
   "Simulating Image-guided Interventions: Interactive Digital Twins to Usher in Next-generation Surgical Suites"

1. LCSR Seminar Series — 04.2023
   Johns Hopkins University, Baltimore, MD
   "An Autonomous X-ray Image Acquisition and Interpretation System for Assisting Percutaneous Pelvic Fracture Fixation"

Selected Press

9. My visit to STIMULATE Research Center in Magdeburg, Germany was featured in a press release.
8. Our work [J-14] was featured in JHU Computer Science News.
7. The Siebel Scholar Class of 2025 was featured in the JHU Hub.
6. Our work [J-9], [J-8], and [J-7] were featured in a JHU Computer Science News article on the IPCAI 2024 conference.
5. Our work in [J-8], [J-5], and [J-3] using the Brainlab Loop-X was featured on JHU Computer Science New.
4. Our work [C-6] presenting the first approach to surgical phase recognition in X-ray guided surgery with dynamic simulation was featured in the JHU Hub and Surgery International.
3. Our work [J-4] demonstrating the utility of synthetic data for training novel X-ray image analysis algorithms was featured in the JHU Engineering magazine, the JHU Hub, and Medical Xpress.
2. My proposal to the Link Fellowship on Simulation, Modeling, and Training was featured on JHU Computer Science News.
1. Our work [J-2] demonstrating efficient strategies for training robots using reinforcement learning was featured in the JHU Hub, TechCrunch, Psychology Today, BBC News, and Voice of America.

Publications

Peer-reviewed Journal Articles

[J-17]. B. Inigo, **B.D. Killeen**, R. Choi, M. Song, A. Uneri, M. Khan, C. Bailey, A. Krieger, M. Unberath. "3D Path Planning for Robot-Assisted Vertebroplasty from Arbitrary Bi-plane X-ray via Differentiable Rendering," *Frontiers in Robotics and AI*, vol. 13, article 1759366, 2026.

[J-16]. F. Klitzner, B. Inigo, **B.D. Killeen**, L. Seenivasan, M. Song, A. Krieger, M. Unberath. "Investigating Robot Control Policy Learning for Autonomous X-ray-guided Spine Procedures." Accepted at *International Journal of Computer Assisted Radiology and Surgery*, 2026.

[J-15]. Y. Sang, Y. Liu, S. Yibulayimu, Y. Wang, **B.D. Killeen**, M. Liu, P.-C. Ku, O. Johannsen, K. Gotkowski, M. Zenk, K. Maier-Hein, F. Isensee, P. Yue, Y. Wang, H. Yu, Z. Pan, Y. He, X. Liang, D. Liu, F. Fan, A. Jurgas, A. Skalski, Y. Ma, J. Yang, S. Płotka, R. Litka, G. Zhu, Y. Song, M. Unberath, M. Armand, D. Ruan, S. K. Zhou, Q. Cao, C. Zhao, X. Wu, Y. Wang. "Benchmark of Segmentation Techniques for Pelvic Fracture in CT and X-ray: Summary of the PENGWIN 2024 Challenge." *Transactions on Medical Imaging*, 2026.

[J-14]. **B.D. Killeen**, A. Suresh, C. Gomez, B. Inigo, C. Bailey, and M. Unberath. "Intelligent Control of Robotic X-ray Devices using a Language-Promptable Digital Twin," *International Journal of Computer Assisted Radiology and Surgery*, 2025.
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2025*
Honored with the **Best Paper Award** at IPCAI'25.

[J-13]. Han Zhang*, **B.D. Killeen***, Y.C. Ku, L. Seenivasan, Y. Zhao, M. Liu, Y. Yang, S. Gu, A. Martin-Gomez, R.H. Taylor, G. Osgood, and M. Unberath. "StraightTrack: Towards Mixed Reality Navigation System for Percutaneous K-wire Insertion," *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*, 2024.
* Joint first authors.
Special Issue: *Augmented Environments for Computer Assisted Interventions (AE-CAI) 2024*

[J-12]. A. Marey, A. Saad, **B.D. Killeen**, C. Gomez, M. Tregubova, M. Unberath, M. Umair. "Gen-AI: Enhancing Patient Education in Cardiovascular Imaging," *BJR|Open*, 2024.

[J-11]. H. Ding, L Seenivasan, **B.D. Killeen**, S.M. Cho, M. Unberath. "Digital Twins as a Unifying Framework for Surgical Data Science: The Enabling Role of Geometric Scene Understanding," *Artificial Intelligence Surgery*, 2024.
Invited submission to the special issue on *Surgical Data Science*

[J-10]. A. Marey, K.C. Serdysnki, **B.D. Killeen**, M. Unberath, M. Umair. "Applications and Implementation of Generative Artificial Intelligence in Cardiovascular Imaging with a Focus on Ethical and Legal Considerations: What Cardiovascular Imagers Need to Know!" *BJR— Artificial Intelligence*, 2024.

[J-9]. **B.D. Killeen***, H. Zhang*, L. Wang, Z. Liu, C. Kleinbeck, M. Rosen, R.H. Taylor, M. Unberath. "Stand in Surgeon's Shoes: Virtual Reality Cross-training to Enhance Teamwork in Surgery," *International Journal of Computer Assisted Radiology and Surgery*, 2024.

Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2024*
Audience vote for **long oral** presentation at IPCAI'24.
Finalist for the **Best Paper Award** at IPCAI'24.

[J-8]. **B.D. Killeen**, S. Chaudhary, G. Osgood, M. Unberath. "Take a Shot! Natural Language Control of Robotic X-ray Systems for Image-guided Surgery," *International Journal of Computer Assisted Radiology and Surgery*, 2024.
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2024*
Audience vote for **long oral** presentation at IPCAI'24.

[J-7]. C. Kleinbeck, H. Zhang, **B.D. Killeen**, D. Roth, M. Unberath. "Neural Digital Twins: Reconstructing Complex Medical Environments for Spatial Planning in Virtual Reality," *International Journal of Computer Assisted Radiology and Surgery*, 2024.
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2024*
Audience vote for **long oral** presentation at IPCAI'24.
Honored with the **Bench-to-Bedside Award** at IPCAI'24.

[J-6]. **B.D. Killeen**, S.M. Cho, M. Armand, R.H. Taylor, M. Unberath. "In Silico Simulation: A Key Enabling Technology for Next-generation Intelligent Surgical Systems," *Progress in Biomedical Engineering*, 2023, vol. 5, no. 3, pp. 032001.
Invited submission to the *Special Issue on In Silico Trials.*

[J-5]. **B.D. Killeen**, C. Gao, K. Oguine, S. Darcy, M. Armand, R.H. Taylor, G. Osgood, M. Unberath. "An Autonomous X-ray Image Acquisition and Interpretation System for Assisting Percutaneous Pelvic Fracture Fixation," *International Journal of Computer Assisted Radiology and Surgery*, 2023.
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2023*
Audience vote for **long oral** presentation at IPCAI'23.
Honorable Mention, **Bench-to-Bedside Award** at IPCAI'23.

[J-4]. C. Gao, **B.D. Killeen**, Y. Hu, R. Grupp, R.H. Taylor, M. Armand, M. Unberath. "Synthetic Data Accelerates the Development of Generalizable Learning-based Algorithms for X-ray Image Analysis," *Nature Machine Intelligence*, 2023, vol. 5, no. 3, pp. 294-308.
Featured in the **JHU Hub**, the **JHU News Letter**, and the **Nature Robotics and AI collection**.

[J-3]. **B.D. Killeen**, J. Winter, W. Gu, A. Martin-Gomez, R.H. Taylor, G. Osgood, M. Unberath. "Mixed Reality Interfaces for Achieving Desired Views with Robotic X-ray Systems," *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*, 2022.
Special Issue: *Augmented Environments for Computer Assisted Interventions (AE-CAI) 2020*

[J-2]. A. Hundt, **B.D. Killeen**, H. Kwon, C. Paxton, G.D. Hager. "'Good Robot!': Efficient Reinforcement Learning for Multi-Step Visual Tasks with Sim to Real Transfer," *IEEE Robotics and Automation Letters*, 2020, vol. 5, no. 4, pp. 6724-6731.
Featured in the **JHU Hub**, **Psychology Today**, **BBC News**, and **Voice of America**.

[J-1]. S. Ambrogio, P. Narayanan, H. Tsai, R. M. Shelby, I. Boybat, C. di Nolfo, S. Sidler, M. Giordano, M. Bodini, N. Farinha, **B.D. Killeen**, C. Cheng, Y. Jaoudi, G. W. Burr. "Equivalent-accuracy accelerated neural-network training using analogue memory," *Nature*, 2018, vol. 558, no. 7708, p. 60.

Peer-reviewed Conference Papers

[C-12]. M. Senne*, **B.D. Killeen***, T. Wang, N. Navab. "Non-intrusive Body Composition Assessment from Full-body mmWave Scans." Accepted at *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026.
*Joint first authors.

[C-11]. M. Wysocki, K.B. Buldu, M. Gafencu, B.D. Killeen, M.F. Azampour, and N. Navab. "OSCAR: Occupancy-based Shape Completion via Acoustic Neural Implicit Representations." Accepted at *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026.

[C-10]. G. Podobnik, N. Balodi, **B.D. Killeen**, T. Vrtovec M. Unberath. "AnatomyGen: Generating Anatomically Plausible Human Phantoms at High Resolution." *International Workshop on Shape in Medical Imaging*, 2025.

[C-9]. **B.D. Killeen**, L.J. Wang, B. Inigo, H. Zhang, M. Armand, R.H. Taylor, G. Osgood, M. Unberath. "FluoroSAM: A Language-promptable Foundation Model for Flexible X-ray Image Segmentation." *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.

[C-8]. M.J. Song, **B.D. Killeen**, J.D. Opfermann, B. Inigo, C. Bailey, A. Uneri, M. Unberath, A. Krieger. "Cannula-mounted Robots for Semi-autonomous Vertebroplasty: A Comparison of Piezo and Screw Inchworm Drive Designs," *International Symposium on Medical Robotics (ISMR)*, 2025.

[C-7]. B. Inigo, Y. Shen, **B.D. Killeen**, M. Song, A. Krieger, C. Bradley, M. Unberath. "An Intrinsically Explainable Approach to Detecting Vertebral Compression Fractures in CT Scans via Neurosymbolic Modeling," *SPIE Medical Imaging*, 2025.

[C-6]. **B.D. Killeen**, H. Zhang, J.E. Mangulabnan, M. Armand, R. Taylor, G. Osgood, M. Unberath. "Pelphix: Surgical Phase Recognition from X-ray Images in Percutaneous Pelvis Fixation," *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023.
Featured in the **JHU Hub** and Surgery International.

[C-5]. **B.D. Killeen**, S. Chakraborty, G. Osgood, **M. Unberath**. "Toward Perception-based Anticipation of Cortical Breach During K-wire Fixation

of the Pelvis," *SPIE Medical Imaging*, 2022.
Selected for **oral** presentation.
Runner up, **Best Paper Award, Physics of Medical Imaging**

[C-4]. J. Opfermann*, **B.D. Killeen***, C. Bailey, M. Khan, A. Uneri, K. Suzuki, M. Armand, F. Hui, A. Krieger†, M. Unberath†. "Feasibility of a Cannula-mounted Piezo Robot for Image-guided Vertebral Augmentation: Toward a Low Cost, Semi-autonomous Approach," *IEEE International Conference on BioInformatics and BioEngineering (BIBE)*, 2021.
* Joint first authors; † joint last authors.
Honored with a **Best Paper Award in Bioengineering**.

[C-3]. X. Liu*, **B.D. Killeen***, A. Sinha, M. Ishii, G. Hager, R. Taylor, M. Unberath. "Neighborhood Normalization for Robust Geometric Feature Learning," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 13049-13058.
* Joint first authors.

[C-2]. C. Gao, X. Liu, W. Gu, **B.D. Killeen**, M. Armand, R. Taylor, M. Unberath. "Generalizing Spatial Transformers to Projective Geometry with Applications to 2D/3D Registration," *Medical Image Computing and Computer Assisted Intervention*, 2020, pp. 329-339.
Code available on **GitHub** here.

[C-1]. X. Liu, Y. Zhang, **B.D. Killeen**, M. Ishii, G. Hager, R. Taylor, M. Unberath. "Extremely Dense Point Correspondences in Multi-view Stereo using a Learned Feature Descriptor," *IEEE Conference on Computer Vision and Pattern Recognition*, 2020, pp. 4847-4856.
Code available on **GitHub** here.

Preprints

[M-9]. F. Tristram, S. Gasperini, B.D. Killeen, M. Walch, C. Benz, N. Navab, G. Ghazaei. "P-JEPA: Procedural Video Representation Learning via Joint Embedding Predictive Architecture." *arXiv preprint*, 2026, arxiv:2606.23256.

[M-8]. S. Rapuri, J. Ko, **B.D. Killeen**, R.H. Taylor, M. Unberath. "2D Versus 3D Diffusion for In Silico Training of Interventional X-ray AI Models." *arXiv preprint*, 2026, arxiv:2606.21414.

[M-7]. M. Senne, **B.D. Killeen**, C. Baur, N. Navab, A. Farshad. "Millimeter-wave Imaging for Anthropometric Body Measurement." *arXiv preprint*, 2026, arxiv:2605.23064.

[M-6]. M. Fehrentz, N. Stellwag, R. Wiebe, N. Thorisch, F. Grob, P. Remerscheid, K.J. Simmoteit, B.D. Killeen, C. Heiliger, N. Navab. "A 4D Representation for Training-Free Agentic Reasoning from Monocular Laparoscopic Video." *arXiv preprint*, 2026, arXiv:2604.00867.

[M-5]. P. Narayanan, S. Ambrogio, C. Mackin, M. Ishii, **B.D. Killeen**, A. Okazaki, K. Hosokawa, J. Luquin, A. Chen, A. Friz, M. Wang, S. Roy, S. Jain, H.

Tsai, G.W. Burr. “Row-by-row Convolutional Neural Networks for Analog-AI.” *Research Square preprint*, 2024, DOI: 10.21203/rs.3.rs-7853227/v1.

[M-4]. H. Zhang, L. Seenivasan, J. L. Porras, R. D. Soberanis-Mukul, H. Ding, H. Shu, **B.D. Killeen**, A. Ghosh, L. Yarmus, M. Ishii, A. C. Argento, M. Unberath. “Did You Just See That? Arbitrary View Synthesis for Egocentric Replay of Operating Room Workflows from Ambient Sensors.” *arXiv preprint*, 2025, arXiv:2510.04802.

[M-3]. **B.D. Killeen***, B. Wan*, A.V. Kulkarni, N. Drenkow, M. Oberst, P.H. Yi, M. Unberath. “Towards Virtual Clinical Trials of Radiology AI with Conditional Generative Modeling.” *arXiv preprint*, 2024, arXiv:2502.09688.

[M-2]. J.Y. Wu*, **B.D. Killeen***, P. Nikutta, M. Thies, A. Zapaishchykova, S. Chakraborty, M. Unberath. “Changes in Reproductive Rate of SARS-CoV-2 Due to Non-pharmaceutical Interventions in 1,417 U.S. Counties,” *medRxiv preprint*, 2020.

[M-1]. **B.D. Killeen**, J.Y. Wu, K. Shah, A. Zapaishchykova, P. Nikutta, A. Tamhane, S. Chakraborty, J. Wei, T. Gao, M. Thies, M. Unberath. “A County-level Dataset for Informing the United States’ Response to COVID-19,” *arXiv preprint*, 2020, arXiv:2004.00756.
The data described herein received a **Kaggle COVID-19 Dataset Award**.
Code available on **GitHub** here.

Patents

[P-2]. **B.D. Killeen**, M. Unberath, “Natural Language Interface for Imaging System.”
US Provisional Application Serial No. TBD, filed December 2023 (pending).

[P-1]. G.W. Burr, **B.D. Killeen**, “Efficient Processing of Convolutional Neural Network Layers Using Analog-memory-based Hardware.”
US Application Serial No. 20200117986, filed March 25, 2019, issued April 16, 2020.

Metadata — This document was last updated on July 2, 2026. A complete, up-to-date version is available at `https://benjamindkilleen.com/files/cv_killeen.pdf`.